# Deep Generative Models I

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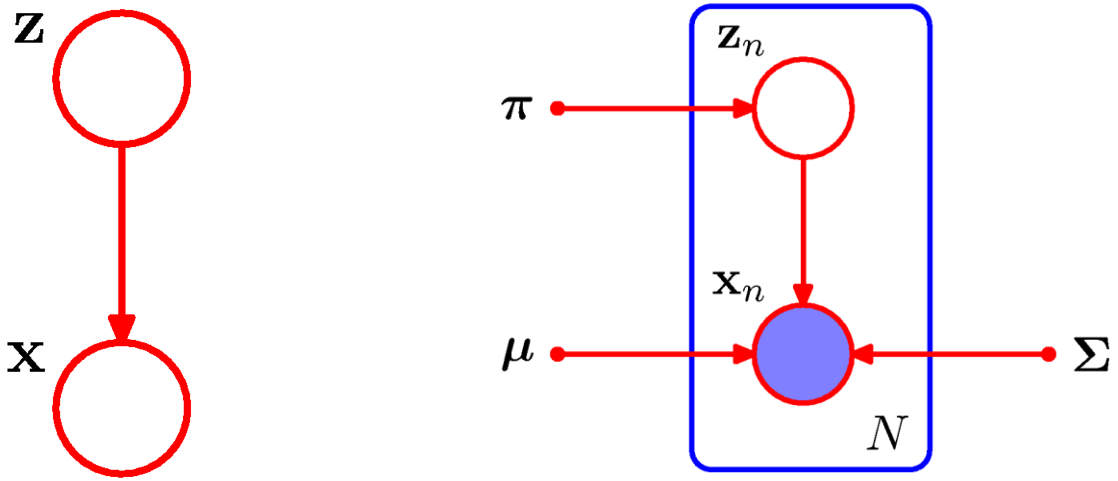
# Outline

- Auto-encoding Variational Bayes
- Generative Adversarial Networks

# GMM with Latent Variables



# Variational Inference

Suppose we have the data X of the model which involves Latent variable Z. The marginal distribution is given as:

$$p(\mathbf{X}|\theta) = \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta)$$

- expressed as summation when the case of Discrete latent variable
- integral for the case of Continuous latent variable

Now, suddenly, suppose we are going to use $\mathrm{q}(Z|\theta')$ , which is the distribution of Z
Then let use consider the following identity expressed with $\mathrm{q}(Z|\theta')$:

$$\ln p(\mathbf{X}|\theta) = L(q, \theta) + KL(q\|p)$$

# Variational Inference

$$\ln p(\mathbf{X}|\theta) = L(q,\theta) + KL(q||p)$$

Where

$$L(q,\theta) = \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X},\mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\}$$

$$KL(q||p) = -\sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X},\theta)}{q(\mathbf{Z})} \right\}$$

(not important)

- the above equations are **functional**s, which take the functions as their domain
  (ex: entropy, KL-divergence, Fourier transform)

- However, L can be represented as $L(\theta; q)$ which is the function of real value of the parameter θ

# Variational Inference

Prior knowledge

- **Jensen's Inequality** (옌슨 부등식)
  - 만약 $f(x)$ 가 `convex` 를 만족한다면 다음이 항상 성립한다.

$$E[f(x)] \geq f(E[x])$$

- **Gibb's Inequality** (깁스 부등식)
  - 함수 $q$ 와 $p$ 에 대해 $KL[p||q] \geq 0$ 을 항상 만족한다.
    - (참고) 만약 $p = q$ 이면 $KL[p||q] = 0$
    - (참고) 만약 $p \neq q$ 이면 $KL[p||q] > 0$
  - (증명) $KL[p||q] = \sum_i p_i \ln \frac{p_i}{q_i} = -\sum_i p_i \ln \frac{q_i}{p_i} \geq -\ln[\sum_i p_i \frac{q_i}{p_i}] = -\ln[\sum_i q_i] = 0$

# Variational Inference

## In the aspect of Lower Bound

$$\ln p(\mathbf{X}|\theta) = \ln \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta)$$

L can be obtained by adopting q(z):

$$\ln \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta) = \ln \sum_{\mathbf{Z}} q(\mathbf{Z}) \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \boxed{\geq} \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\} = L(q, \theta)$$

**Jensen's Inequality**

The above $L(\theta; q)$ is the lower bound of log likelihood $\ln p(\mathbf{X}|\theta)$
(**E**vidence **L**ower **BO**und, ELBO)

# Variational Inference

## In the aspect of Lower Bound

Proof) let us compute the difference btwn log likelihood $\ln p(X|\theta)$ and $L(\theta; q)$:

$$\ln p(\mathbf{X}|\theta) - L(q,\theta) = \ln p(\mathbf{X}|\theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln\left\{\frac{p(\mathbf{X},\mathbf{Z}|\theta)}{q(\mathbf{Z})}\right\}$$

$$= \ln p(\mathbf{X}|\theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln\left\{\frac{p(\mathbf{Z}|\mathbf{X},\theta)p(\mathbf{X}|\theta)}{q(\mathbf{Z})}\right\}$$

$$= \ln p(\mathbf{X}|\theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln\left\{\frac{p(\mathbf{Z}|\mathbf{X},\theta)}{q(\mathbf{Z})}\right\} - \ln p(\mathbf{X}|\theta)\sum_{\mathbf{Z}} q(\mathbf{Z})$$

$$= -\sum_{\mathbf{Z}} q(\mathbf{Z}) \ln\left\{\frac{p(\mathbf{Z}|\mathbf{X},\theta)}{q(\mathbf{Z})}\right\} = KL[q(\mathbf{Z})||p(\mathbf{Z}|\mathbf{X},\theta)] = KL[q||p]$$

finally we have:

$$\ln p(\mathbf{X}|\theta) = L(q,\theta) + KL(q||p) \boxed{\geq} L(\theta; q)$$

**Gibb's Inequality**

# Variational Inference

**In the aspect of Posterior Approximation**

$$KL(q(z)||p(z|x)) = \int q(z) \ln \frac{q(z)}{p(z|x)} dz$$

$$= \int q(z) \ln \frac{q(z)p(x)}{p(x|z)} dz$$

$$= \int q(z) \ln p(x)\, dz + \int q(z) \ln \frac{q(z)}{p(x,z)} dz$$

$$= \ln p(x) - L(\theta; q) \boxed{\geq} 0$$

**Gibb's Inequality**

# Variational Inference

$$\ln p(x) = L(\theta; q) + KL(q(z)||p(z|x,\theta))$$

- We want to find proper $q(z)$ and $\theta$ that maximize target distribution $\ln p(x)$
- Hard to find optimal solution simultaneously
- Use Expectation-Maximization (find $q(z)$ and $\theta$ iteratively)

We have two issues:

1) how to find $q(z)$ and $\theta$ simultaneously?
2) find $q(z)$ in the form of more complex function (related to $x$)

# Variational Inference

## Evidence Lower Bound

$$\log p_\theta(x) = KL(q_\phi(z)||p_\theta(z|x)) + L(\theta, \phi)$$

$$\log p_\theta(x) \geq L(\theta, \phi)$$

We want to maximize lower bound in order to find $p_\theta(x)$

In GMM case using Expectation-Maximization,
we simply choose $q_\phi(z)\ as\ p_\theta(z|x)$
What if the posterior is intractable, thereby impossible to choose?

# Variational Inference

**Evidence Lower Bound**

$$\boxed{\log p_\theta(x) \geq L(\theta, \phi)}$$

$$L(\theta, \phi)$$

$$= \int q_\phi(z) \log \frac{p_\theta(x, z)}{q_\phi(z)} dz$$

$$= \int q_\phi(z) \log \frac{p_\theta(z) p_\theta(x|z)}{q_\phi(z)} dz$$

$$= \int q_\phi(z) \log \frac{p_\theta(z)}{q_\phi(z)} dz + \int q_\phi(z) \log p_\theta(x|z)\, dz$$

$$= -KL\big(q_\phi(z) \big| p_\theta(z)\big) + \; E_{z \sim q_\phi}[\log p_\theta(x|z)]$$

# Variational Inference

**Evidence Lower Bound**

$$\boxed{\log p_\theta(x) \geq L(\theta,\phi)}$$

$$L(\theta,\phi) = -KL\big(q_\phi(z)\big|p_\theta(z)\big) + E_{z\sim q_\phi}[\log p_\theta(x|z)]$$

We have two issues:

1) how to find $q(z)$(in our case $\phi$) and $\theta$ simultaneously?

# Variational Inference

**Evidence Lower Bound**

$$\boxed{\log p_\theta(x) \geq L(\theta, \phi)}$$

$$L(\theta, \phi)$$
$$= -KL\big(q_\phi(z) \big| p_\theta(z)\big) + E_{z \sim q_\phi}[\log p_\theta(x|z)]$$

We have two issues:

1) how to find $q(z)$(in our case $\phi$) and $\theta$ simultaneously?

- To relax the first issue, We want to **differentiate** and optimize the lower bound w.r.t. both the variational parameters $\phi$ and generative parameters $\theta$.

# Variational Inference

**Evidence Lower Bound**

$$\boxed{\log p_\theta(x) \geq L(\theta,\phi)}$$

$$L(\theta,\phi) \\ = -KL\big(q_\phi(z)\big|p_\theta(z)\big) + \boxed{E_{z\sim q_\phi}[\log p_\theta(x|z)]}$$

We have two issues:

**1) how to find $\boldsymbol{q(z)}$(in our case $\boldsymbol{\phi}$) and $\boldsymbol{\theta}$ simultaneously?**

- To relax the first issue, We want to **differentiate** and optimize the lower bound w.r.t. both the variational parameters $\phi$ and generative parameters $\theta$.

# Variational Inference

**Evidence Lower Bound**

$$\boxed{\log p_\theta(x) \geq L(\theta, \phi)}$$

$$E_{z\sim q_\phi}[\log p_\theta(x|z)]$$

The usual (na¨ıve) Monte Carlo gradient estimator for this type of problem is:

$$E_{z\sim q_\phi}[\log p_\theta(x|z)] \cong \frac{1}{N}\sum_n \log p_\theta(x|z^n)$$

where

$$z^n \sim q_\phi(z)$$

# Variational Inference

**Evidence Lower Bound**

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where

$$z^n \sim q_\phi(z)$$

$$\boxed{\begin{aligned} &E_{z\sim q_\phi}[\log p_\theta(x|z)] \\ &= \int q_\phi(z) \log p_\theta(x|z)\, dz \end{aligned}}$$

$$=> \quad \nabla E_{z\sim q_\phi}[\log p_\theta(x|z)] \cong \frac{1}{N}\sum_n \nabla \log p_\theta(x|z^n)$$

# Variational Inference

## Evidence Lower Bound

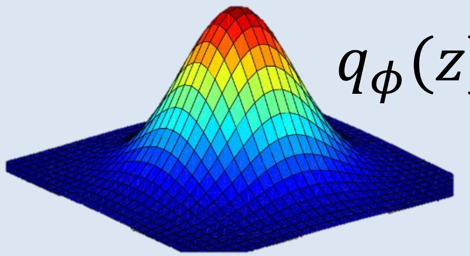
$$\log p_\theta(x) \geq L(\theta, \phi)$$

$$L(\theta, \phi)$$

$$= -KL\big(q_\phi(z) \big| p_\theta(z)\big) + \frac{1}{N} \boxed{\sum_n \log p_\theta(x|z^n)}$$

where

$$\boxed{z^n \sim q_\phi(z)}$$



$q_\phi(z)$

Sampling $\xrightarrow{z^n}$ $p_\theta(x^i | z^n)$ $\Rightarrow$ $x^i$

# Variational Inference

**Evidence Lower Bound**

$$\boxed{\log p_\theta(x) \geq L(\theta, \phi)}$$

$$E_{z \sim q_\phi}[\log p_\theta(x|z)]$$

We have two issues:

**2) find $q(z)$ in the form of more complex function (related to $x$)**

For $q(z)$, we don't have any constraints of it; we can define $q(z)$ as we want:

$$q_\phi \triangleq q_\phi(z|x)$$ **Variational Likelihood**

Ex) $N\left(z;\ \mu_\phi(x^i), \sigma_\phi(x^i)^2 I\right)$

# Variational Inference

**Evidence Lower Bound**

$$\boxed{\log p_\theta(x) \geq L(\theta, \phi)}$$

$$L(\theta, \phi)$$

$$= -KL\big(q_\phi(z|x)\big|p_\theta(z)\big) + \boxed{\frac{1}{N}\sum_n \log p_\theta(x|z^n)}$$

where

$$\boxed{z^n \sim q_\phi(z|x)}$$

# Variational Inference

## Evidence Lower Bound

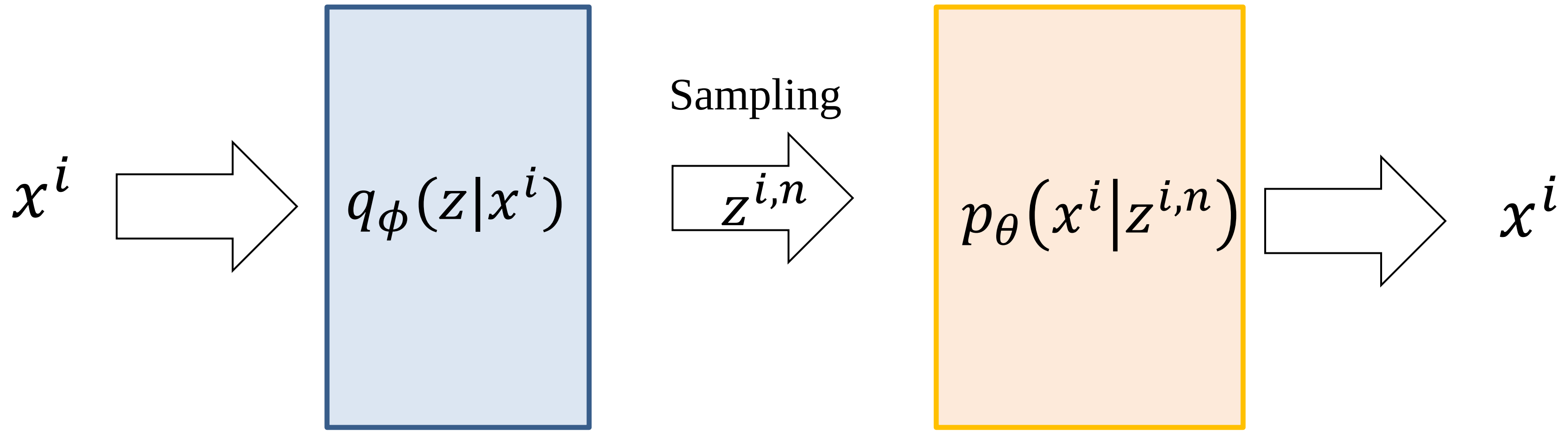
$$\boxed{\log p_\theta(x) \geq L(\theta, \phi)}$$

$$L(\theta, \phi)$$

$$= -KL\big(q_\phi(z|x)\big|p_\theta(z)\big) + \boxed{\frac{1}{N}\sum_n \log p_\theta(x|z^n)}$$

where

$$\boxed{z^n \sim q_\phi(z|x)}$$



# Variational Inference

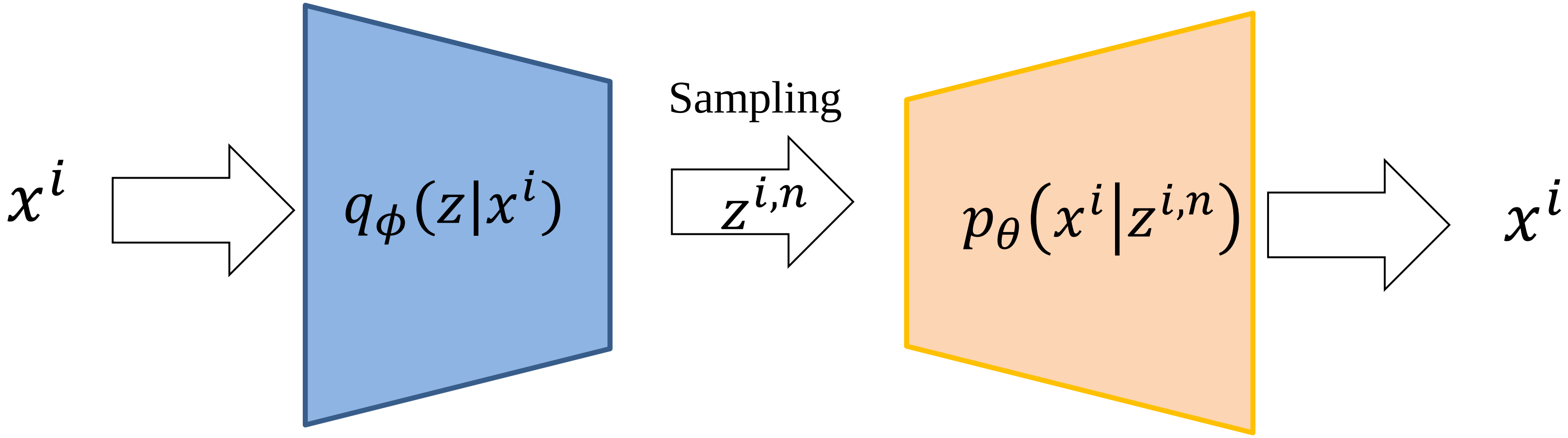
**Evidence Lower Bound**

$$\log p_\theta(x) \geq L(\theta, \phi)$$

$$L(\theta, \phi)$$
$$= -KL\big(q_\phi(z|x) \big| p_\theta(z)\big) + \boxed{\frac{1}{N}\sum_n \log p_\theta(x|z^n)}$$

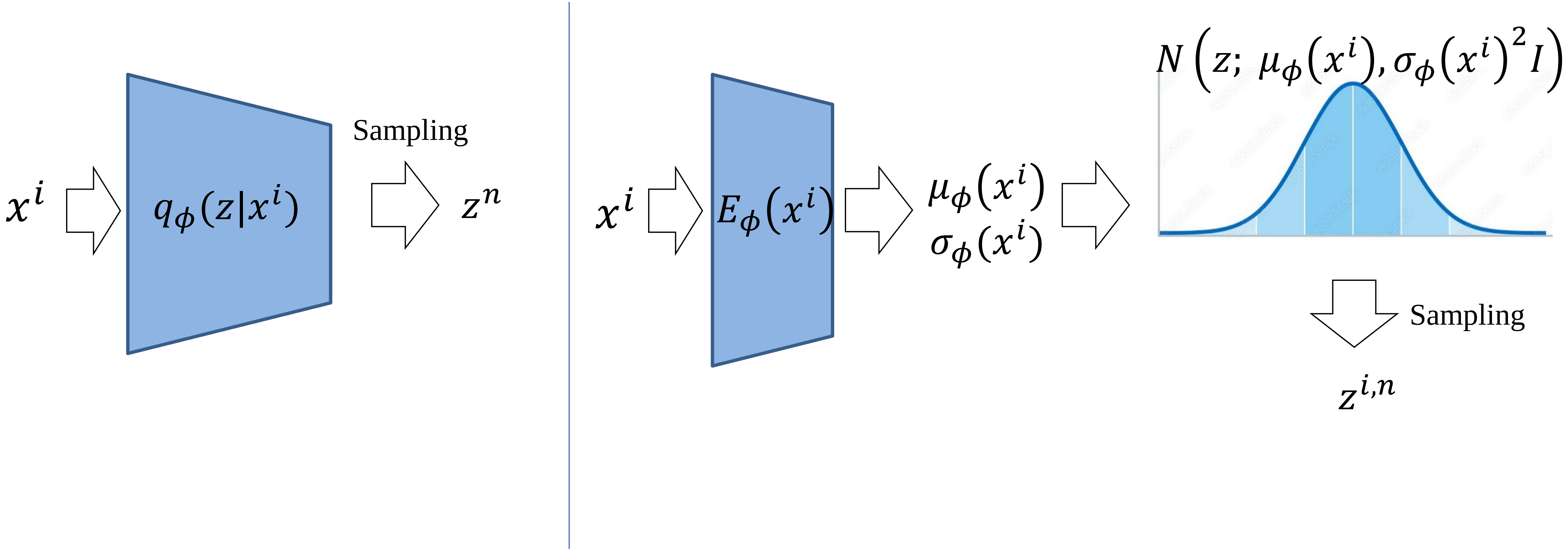
where

$$\boxed{z^n \sim q_\phi(z|x)}$$



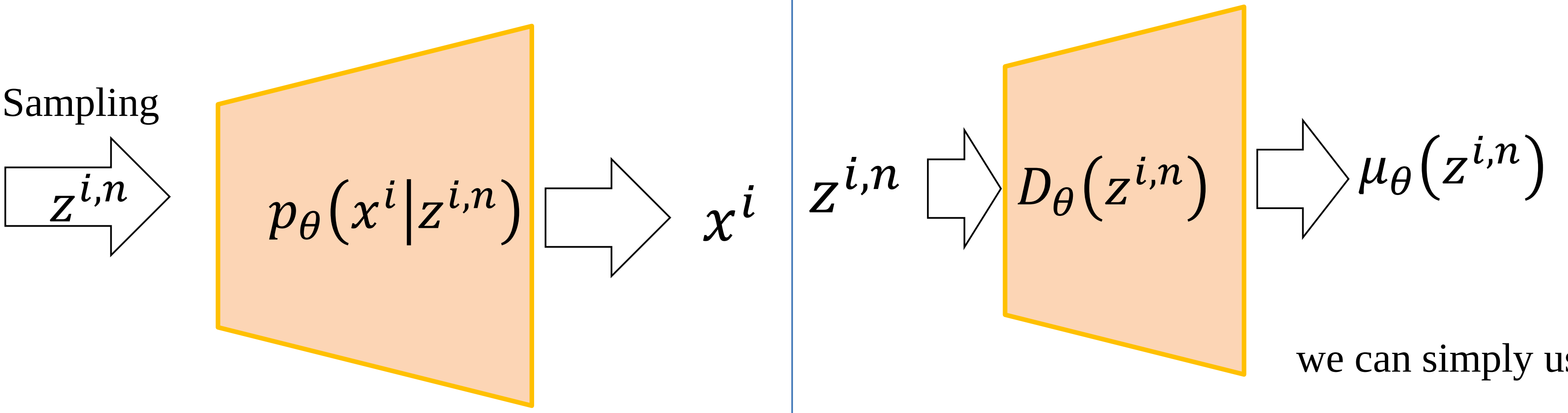
# Variational Inference

$$q_\phi(z|x) \triangleq N\left(z;\, \mu_\phi(x), \sigma_\phi(x)^2 I\right)$$



# Variational Inference

Ex) image reconstruction $$p_\theta\left(x^i \middle| z^{i,n}\right) \triangleq N\left(x;\ \mu_\theta\left(z^{i,n}\right), \sigma^2 I\right)$$



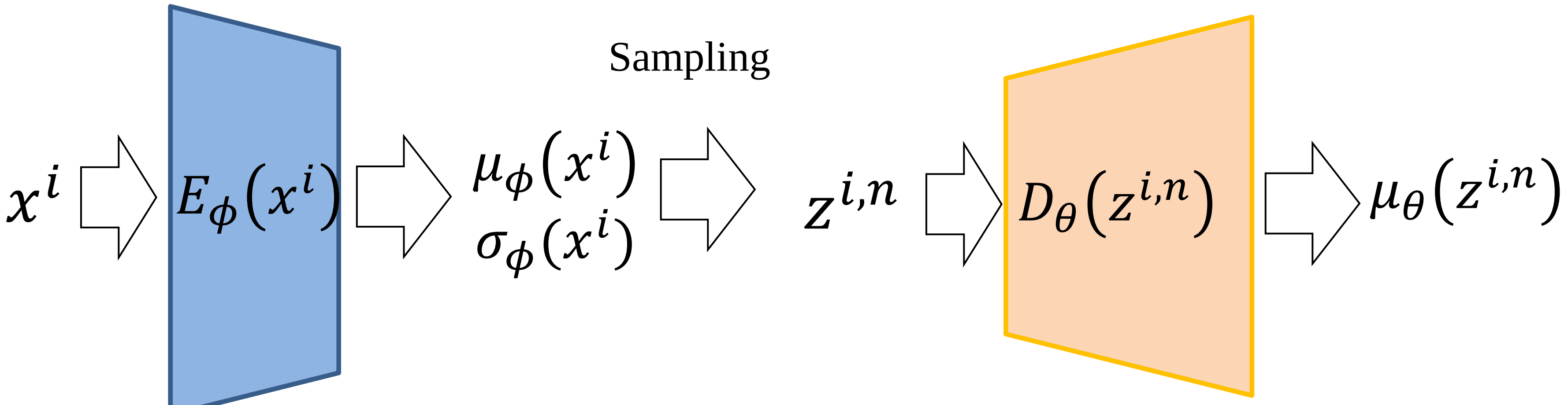

we can simply use MSE, by letting $\sigma$ be a constant:

$$\frac{1}{N}\sum_n \left|\mu_\theta\left(z^{i,n}\right) - x^i\right|^2$$

$$= \frac{1}{N}\sum_n \left|D_\theta\left(z^{i,n}\right) - x^i\right|^2$$

# Variational Inference

$$L(\theta,\phi) = -KL\big(q_\phi(z|x)\big|p_\theta(z)\big) + \boxed{\frac{1}{N}\sum_n \log p_\theta(x|z^n)}$$

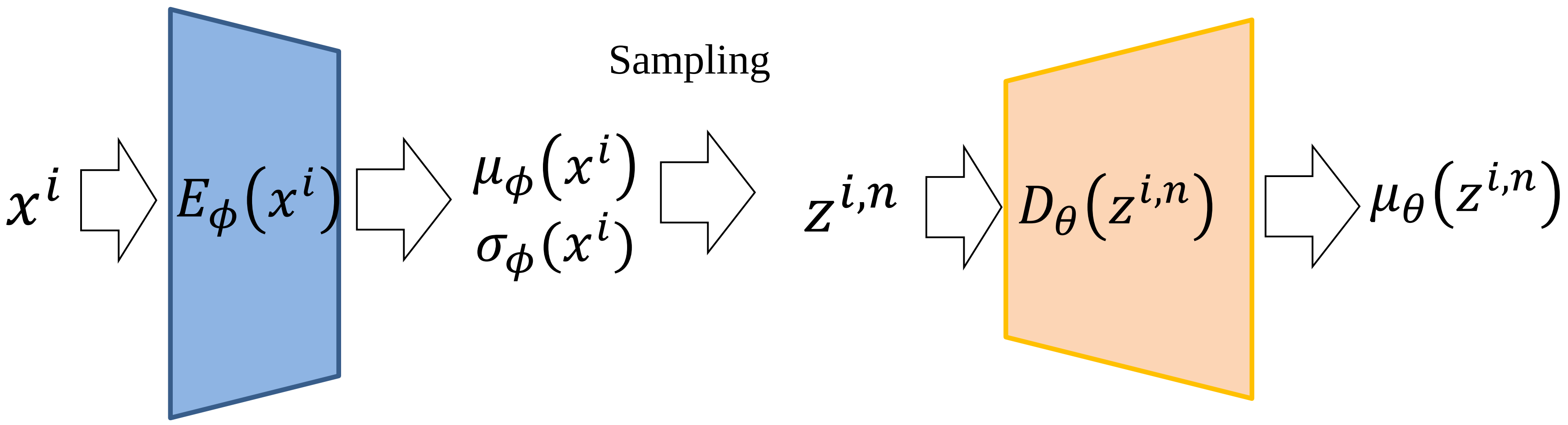



For loss, we can simply use MSE:

$$\frac{1}{N}\sum_n |\mu_\theta(z^{i,n}) - x^i|^2 \quad = \frac{1}{N}\sum_n |D_\theta(z^{i,n}) - x^i|^2$$

# Variational Inference

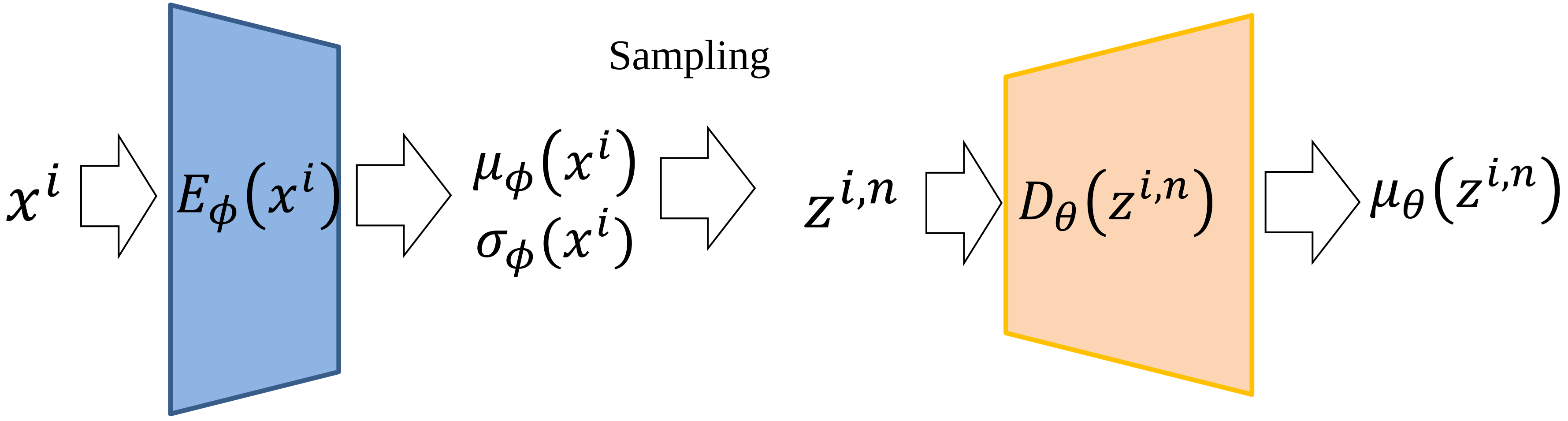
$$L(\theta,\phi) = \boxed{-KL\left(q_\phi(z|x)\middle|p_\theta(z)\right)} + \frac{1}{N}\sum_n \log p_\theta(x|z^n)$$



# Variational Inference

$$L(\theta,\phi) = \boxed{-KL\big(q_\phi(z|x)\big|p_\theta(z)\big)} + \frac{1}{N}\sum_n \log p_\theta(x|z^n)$$

Constraints for
latent variables



# Variational Inference

$$L(\theta,\phi) = \boxed{-KL\left(q_\phi(z|x) \middle| p_\theta(z)\right)} + \frac{1}{N}\sum_n \log p_\theta(x|z^n)$$

Constraints for
latent variables

$$q_\phi(z|x) \triangleq N\left(z;\ \mu_\phi(x), \sigma_\phi(x)^2 I\right)$$

$$p_\theta(z) \triangleq N(z; 0, I)$$

# Variational Inference

$$L(\theta,\phi) = \boxed{-KL\big(q_\phi(z|x)\big|p_\theta(z)\big)} + \frac{1}{N}\sum_n \log p_\theta(x|z^n)$$

$$\int q_{\boldsymbol{\theta}}(\mathbf{z}) \log p(\mathbf{z})\, d\mathbf{z} = \int \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \boldsymbol{\sigma}^2) \log \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})\, d\mathbf{z}$$

$$= -\frac{J}{2}\log(2\pi) - \frac{1}{2}\sum_{j=1}^{J}(\mu_j^2 + \sigma_j^2)$$

$$\int q_{\boldsymbol{\theta}}(\mathbf{z}) \log q_{\boldsymbol{\theta}}(\mathbf{z})\, d\mathbf{z} = \int \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \boldsymbol{\sigma}^2) \log \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \boldsymbol{\sigma}^2)\, d\mathbf{z}$$

$$= -\frac{J}{2}\log(2\pi) - \frac{1}{2}\sum_{j=1}^{J}(1 + \log \sigma_j^2)$$

# Variational Inference

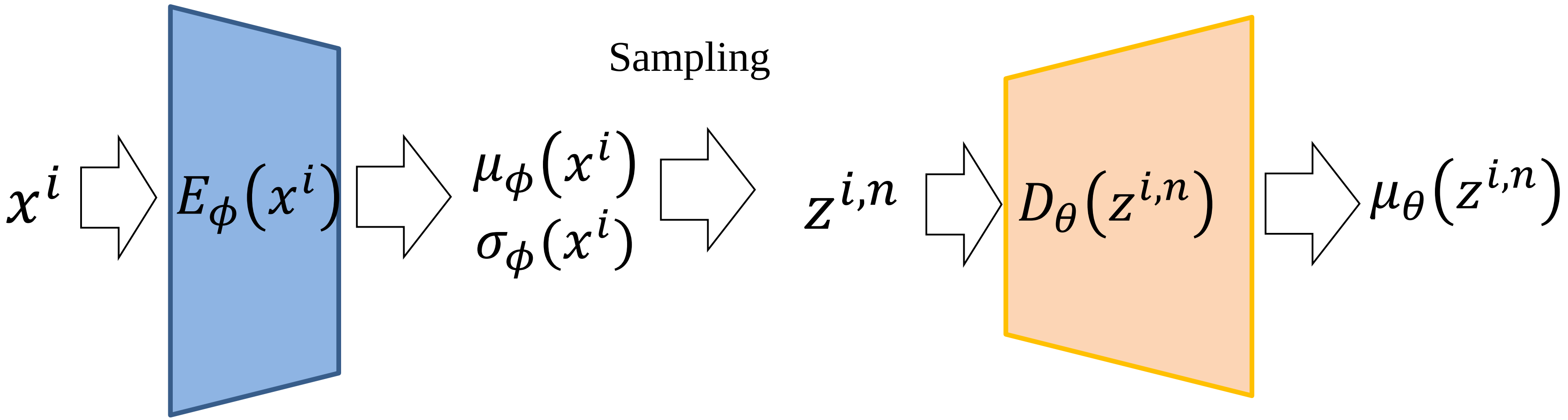
$$L(\theta,\phi) = \boxed{-KL\big(q_\phi(z|x)\big|p_\theta(z)\big)} + \frac{1}{N}\sum_n \log p_\theta(x|z^n)$$

$$-KL\big(q_\phi(z|x)\big|p_\theta(z)\big) = \int q_{\boldsymbol{\theta}}(\mathbf{z})\left(\log p_{\boldsymbol{\theta}}(\mathbf{z}) - \log q_{\boldsymbol{\theta}}(\mathbf{z})\right) d\mathbf{z}$$

$$= \frac{1}{2}\sum_{j=1}^{J}\left(1 + \log((\sigma_j)^2) - (\mu_j)^2 - (\sigma_j)^2\right)$$

# Variational Inference

$$L(\theta,\phi) = -KL\left(q_\phi(z|x)\middle|p_\theta(z)\right) + \boxed{\frac{1}{N}\sum_n \log p_\theta(x|z^n)}$$




For loss, we can simply use MSE + KL-constraint term:

$$\frac{1}{N}\sum_n |D_\theta(z^{i,n}) - x^i|^2 + \frac{1}{2}\sum_{j=1}^{J}(1 + \log((\sigma_j{}^2)) - \left(\mu_j\right)^2 - (\sigma_j{}^2)$$

# Variational Inference

Now we have:

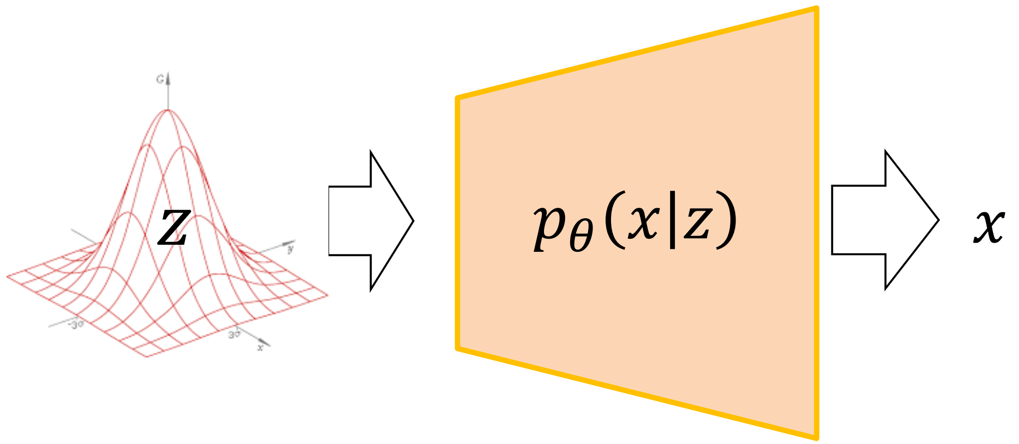
1) likelihood $p_\theta(x|z)$ (Decoder)
2) posterior $p_\theta(z|x) \cong q_\phi(z|x)$ (Encoder)
3) prior $p_\theta(z)$ (by assumption)

- Bayesian decision

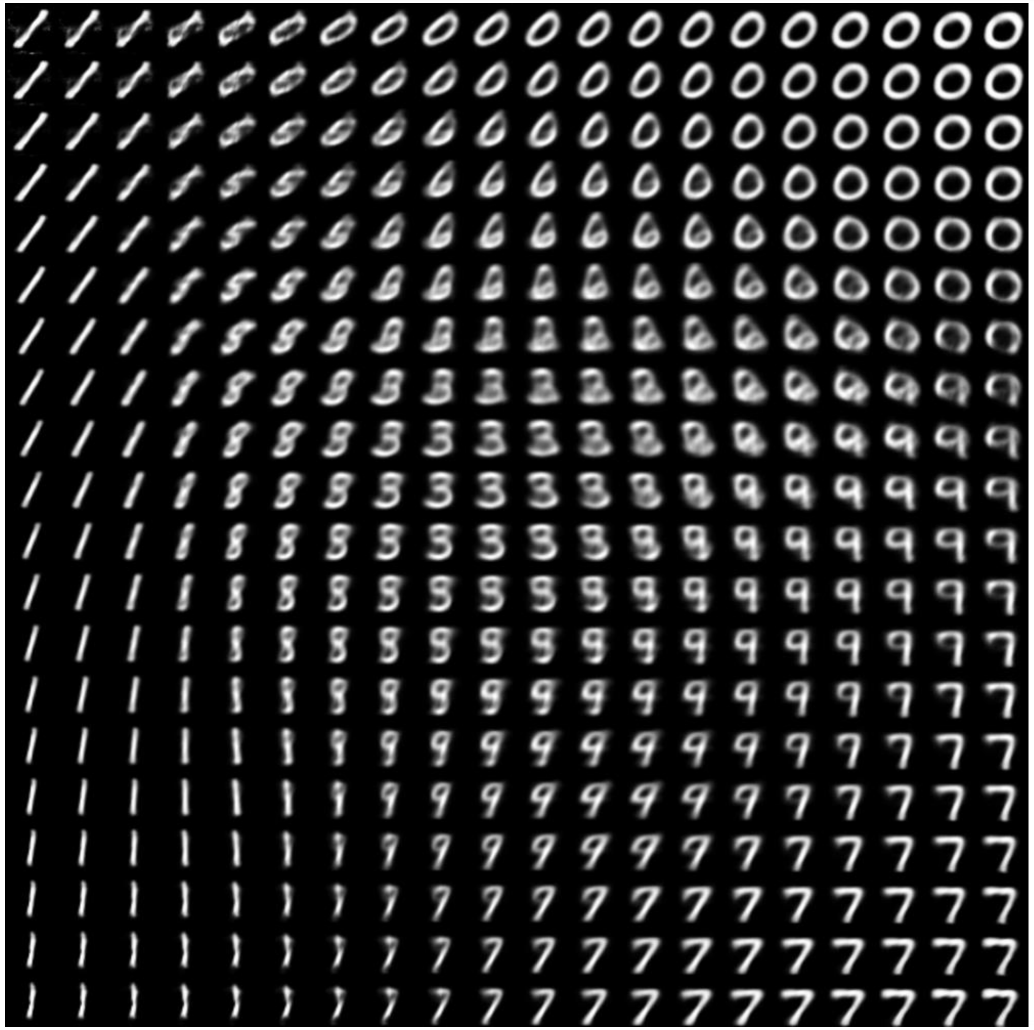
- Generative model

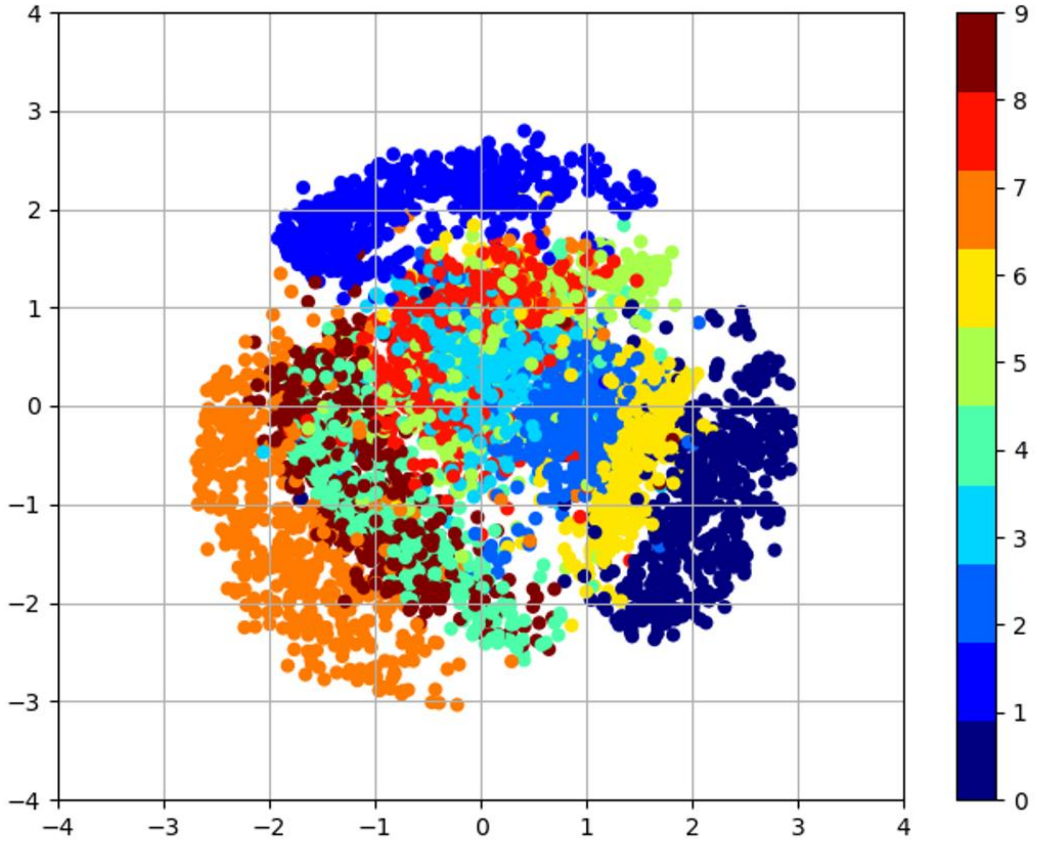
$$x \sim p_\theta(x) = \int p_\theta(x|z)p(z)dz$$

$$x \sim p_\theta(x|z) \text{ where } z \sim p_\theta(z)$$



# Variational Inference






$x \sim p_\theta(x|z)$ where $z \sim p_\theta(z)$

$z \sim q_\phi(z|x)$

# Interim Summary

- Auto-encoding Variational Bayes
- Generative Adversarial Networks